

DSA QUIZ APPLICATION

Minor Project Report • Android App Development + Java

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Language: Java

1. Project Objective

The objective of this project is to develop a native, interactive Android application in **Java** that helps students and developers test, practice, and sharpen their **Data Structures and Algorithms (DSA)** knowledge. The app provides a timed quiz environment, instant solution feedback, scoring metrics, answer review capabilities, and local persistence for session tracking and leaderboards.

2. Project Description

DSA Quiz is a modern, pastel-themed native Android application designed to make learning algorithms engaging and accessible. Learners can select specific DSA topics (such as Arrays, Linked Lists, Stacks & Queues, Trees, Graphs, Sorting, Dynamic Programming, and Time Complexity) and choose appropriate difficulty levels (Basic, Medium, Hard).

The application features a 30-second per-question countdown timer, a 50:50 lifeline feature, instant answer checking with color-coded feedback and technical explanations, detailed session scoring, a question review screen, a local leaderboard podium, and lifetime progress statistics.

3. Key Features Implemented

- **Multi-Topic DSA Selection:** Support for 8 core DSA domains (Arrays, LinkedLists, Stacks/Queues, Trees, Graphs, Sorting, DP, Time Complexity).
- **Difficulty Grading:** Basic, Medium, and Hard difficulty levels.
- **Timed Quiz Challenge Mode:** Per-question 30-second countdown timer with animated progress bar and automatic timeout handling.
- **Untimed Practice Mode:** Stress-free study mode for conceptual understanding without time constraints.
- **50:50 Lifeline:** Eliminates two incorrect options to assist the user.
- **Instant Solution & Explanation:** Color-coded card feedback (Green for Correct, Red for Incorrect) with in-depth technical explanations.
- **No-Repeat Question Logic:** Ensures questions within an active session are not repeated.

- **Result & Performance Summary:** Displays score, accuracy percentage, time taken, streak counter, and achievement badges.
- **Review Answers Screen:** Enables users to examine all questions, their chosen answers vs correct answers, and corresponding explanations.
- **Top 3 Leaderboard Podium & Ranking:** Displays top scores with gold, silver, and bronze podium cards.
- **Statistics & Mastery Progress:** Tracks total points, quizzes completed, daily streak, and topic mastery meters.
- **DSA Study Cheat Sheet:** Quick reference guides for key algorithms, time complexities, and properties.
- **In-App Project Report:** Integrated documentation directly accessible from the Home Dashboard.

4. Technologies & Tools Used

Java (JDK 11 / 17)

Android Studio

Room Database (SQLite)

Material Components

RecyclerView

Git & GitHub

5. Android Components Used

Component	Role in Application
Activity	Manages screen lifecycles for Splash, Home, Topic Selection, Quiz, Results, Review, Leaderboard, Stats, Study, and Report.
Room Database	Stores question entities, user quiz history records, and leaderboard rankings offline with SQLite ORM.
RecyclerView	Renders dynamic, smooth lists for answer reviews and leaderboard rankings.
CountDownTimer	Executes real-time 30-second countdowns for each question in Challenge mode.
Intent	Passes selected topic, difficulty, scores, and serialized question collections across activities.
SharedPreferences	Stores user points, day streaks, user profile name, and rotating tip indices.
MaterialCardView	Provides modern, rounded pastel cards with soft elevations and responsive states.

6. Project Architecture & Structure

```
com.example.quizapp
├── activities/      (SplashActivity, HomeActivity, TopicSelectionActivity,
│                   QuizActivity, ResultActivity, ReviewAnswersActivity,
│                   LeaderboardActivity, StatsActivity, StudyActivity, ReportActivity)
├── adapters/       (LeaderboardAdapter, ReviewAnswersAdapter)
├── database/       (AppDatabase, QuestionDao, HistoryDao, LeaderboardDao, DataSeeder)
├── models/         (Question, QuizHistory, LeaderboardEntry, QuizSessionItem)
└── utils/         (DsaTips, PreferenceManager)
```

7. Challenges Faced & Solutions

- Preventing Question Repetition:** Implemented in-memory tracking with Room query filters to guarantee variety per session.
- Smooth State & Timer Management:** Handled proper lifecycle termination of `CountDownTimer` on activity transition to prevent memory leaks.
- Dynamic Feedback Banner:** Developed adaptable `MaterialCardView` widgets that update background tint, border, and explanation content instantly upon submission.

8. Key Learning Outcomes

- Deepened understanding of native Android development using **Java** and Android SDK.
 - Gained hands-on proficiency with **Room Database (SQLite ORM)** for offline persistence.
 - Mastered **RecyclerView** and custom adapters for complex dynamic list rendering.
 - Learned clean UI/UX design principles utilizing pastel color schemes and Material Design standards.
 - Practiced professional Git version control workflows and progressive commit history.
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